

1 Locally Convex Spaces

We introduce some basic facts about locally convex space, i.e, topological vector space whose topologies are defined by a family of seminorms.

Seminorms

Definition 1.1. : Let k be the field $\mathbb{R}$ or $\mathbb{C}$ and V is a k -vector space. A **seminorm** on V is a mapping $p : V \rightarrow [0, +\infty)$ such that

- (1) The mapping p is **positively homogenous**, i.e., $p(\alpha x) = |\alpha|p(x)$ for all $\alpha \in k$ and all $x \in V$.
- (2) The mapping p is **subadditive**, i.e., $p(x + y) \leq p(x) + p(y)$ for all $x, y \in V$.

Example 1.1. : Any norm $\|\cdot\|$ on V is certainly a seminorm.

Example 1.2. : Take $V = \mathcal{R}[a, b]$, the space of Riemann-integrable functions on the closed interval $[a, b]$ and define

$$p : \mathcal{R}[a, b] \rightarrow \mathbb{R}$$

$$f \mapsto \int_a^b |f(t)| dt$$

Then it's easy to see that p is a seminorm on $V = \mathcal{R}[a, b]$. But p is not a norm since $p(f) = \int_a^b |f(t)| dt = 0$ does not imply that $f = 0$.

In contrast to the topologies induced by norms, topologies induced by seminorms need not to be Hausdorff.

Lemma 1.1. : Let I be an index set, V is a vector space over k and $(p_i)_{i \in I}$ is a family of seminorms on V . For a finite subset $F \subset I$ and $\varepsilon > 0$, define

$$U_{F;\varepsilon} := \bigcap_{i \in F} \{x \in V \mid p_i(x) < \varepsilon\}$$

and define

$$\mathfrak{U} := \{U_{F;\varepsilon} \mid \varepsilon > 0, F \subset I, \#F < \infty\}.$$

Then the set

$$\tau := \{O \subseteq V \mid \text{For all } x \in V \text{ there is some } U \in \mathfrak{U} \text{ such that } x + U \subseteq O\}$$

defines a topology on V .

Proof: It's clear that $V \subseteq \mathfrak{U}$ and $\phi \subseteq \mathfrak{U}$.

Let J be an index set and $(O_j)_{j \in J}$ be an arbitrary family in τ . Let $O := \bigcup_{j \in J} O_j$ and let $x \in O$. Then $x \in O_j$ for some $j \in J$. Since $O_j \in \tau$, so there exists some $U \in \mathfrak{U}$ such that $x + U \subseteq O_j \subseteq O$, which means that $O = \bigcup_{j \in J} O_j \in \tau$;

Let $O_1, O_2, \dots, O_n \in \tau$. Let $O := \bigcap_{j=1}^n O_j$ and let $x \in O$. Since $O_j \in \mathfrak{U}$, there are $U_j \in \mathfrak{U}$ such that $x + U_j \subseteq O_j$ ($j = 1, 2, \dots, n$). Since every U_j is of the form $U_j = U_{F_j; \varepsilon_j}$ where F_j is a finite subset of I and $\varepsilon_j > 0$. Now define $\varepsilon := \min_{1 \leq j \leq n} \varepsilon_j$ and $F := \bigcup_{j=1}^n F_j$, hence F is a finite subset of I and $\varepsilon > 0$. Thus we can know that $U_{F; \varepsilon} \in \mathfrak{U}$ and $U_{F; \varepsilon} \subseteq \bigcap_{j=1}^n U_j$, hence

$$x + U_{F; \varepsilon} \subseteq \bigcap_{j=1}^n (x + U_j) \subseteq O,$$

which shows that $O = \bigcap_{j=1}^n O_j \in \tau$. $\square$

Remark: A vector space with a topology induced by a family of seminorms as above is called a **locally convex space**. The topology τ turns V into a topological vector space, i.e., vector addition and multiplication with scalars are continuous mappings with respect to this topology.

Remark: Topological vector spaces whose topologies can be induced by a metric are called **metrizable**. If the topology can be induced by a norm, then it is called **normable**.

In the abstract setting of general locally convex spaces, the sets $U_{F; \varepsilon}$ play the role of the open ε -balls $B_\varepsilon^n(0)$ in $\mathbb{C}^n$ or $\mathbb{R}^n$. i.e., they form a basis of open neighborhoods of zero.

Convergence in Locally Convex Space

Definition 1.2. : Let $(V, (p_i)_{i \in I})$ be a locally convex space, τ is the topology induced by the family of seminorms $(p_i)_{i \in I}$, $\mathfrak{U}$ is the basis of neighborhoods of zeros as defined in the lemma above.

- (1) A sequence $(x_j)_{j \in \mathbb{N}}$ in V **converges to the limit** $x \in V$ if for every $U \in \mathfrak{U}$, there is some $N = N_U \in \mathbb{N}$ such that $x - x_j \in U$ for all $j \geq N$.
- (2) The sequence $(x_j)_{j \in \mathbb{N}}$ is called **Cauchy sequence** if for every $U \in \mathfrak{U}$, there is some $N = N_U \in \mathbb{N}$ such that $x_k - x_l \in U$ for all $k, l \geq N$.
- (3) The space V is called **sequentially complete** with respect to the topology τ if every Cauchy sequence converges in V .

Lemma 1.2. : A sequence $(x_j)_{j \in \mathbb{N}}$ converges to $x \in V \Leftrightarrow \lim_{j \rightarrow \infty} p_i(x_j - x) = 0$ for all $i \in I$.

Proof: By the definition of $\mathfrak{U}$ and the convergence of $(x_j)_{j \in \mathbb{N}}$ in V . $\square$

2 The Compact-Open Topology on $\mathcal{C}(U, E)$

Lemma 2.1. : Let $U \subseteq \mathbb{C}^n$ be open. For every $j \in \mathbb{N}_+$, define

$$K_j := \{z \in U \mid \|z\|_\infty \leq j, \text{dist}_{\|\cdot\|_\infty}(z, \mathbb{C}^n \setminus U) \geq \frac{1}{j}\}.$$

Then we have:

- (1) Every K_j is a compact set;
- (2) For all $j \geq 1$, we have $K_j \subseteq K_{j+1}^\circ$;
- (3) The set U is the union of K_j , i.e., $U = \bigcup_{j=1}^\infty K_j$;
- (4) If K is an arbitrary compact subset of U , then there is some $j_K \in \mathbb{N}_+$ such that $K \subseteq K_{j_K}$.

Proof:(1) By the definition of K_j , K_j is an intersection of two closed sets and $K_j \subseteq B_j^n(0)$, so by Heine-Borel Theorem, we know that K_j is compact;

(2) Since the set

$$\{z \in U \mid \|z\|_\infty < j+1, \text{dist}_{\|\cdot\|_\infty}(z, \mathbb{C}^n \setminus U) > \frac{1}{j+1}\}$$

is an open set and contains K_j , hence $K_j \subseteq K_{j+1}^\circ$;

(3) If $U = \emptyset$, the conclusion clearly holds. Let $U \neq \emptyset$, since U is open, for $\forall z \in U$, then z has a positive distance to the complement of U , i.e., $\exists j_1 \in \mathbb{N}_+$ such that

$$\text{dist}_{\|\cdot\|_\infty}(z, \mathbb{C}^n \setminus U) \geq \frac{1}{j_1}$$

Since $z \in U$, $\exists j_2 \in \mathbb{N}_+$ such that $\|z\|_\infty \leq j_2$. Take $j_3 = \max\{j_1, j_2\}$, hence $z \in K_{j_3}$, which implies that $U \subseteq \bigcup_{j=1}^\infty K_j$. Since K_j is a subset of U , so $U = \bigcup_{j=1}^\infty K_j$;

(4) Let K be a compact subset of U , then the sets $\{K_j^\circ \mid j \in \mathbb{N}_+\}$ is an open cover of K . So $\exists j_K \in \mathbb{N}_+$ such that

$$K \subseteq \bigcup_{i=1}^{j_K} K_i$$

since we know that $K_j \subseteq K_{j+1}$, thus $K \subseteq K_{j_K}$. $\square$

Remark: A sequence of compact sets having the properties 2 and 3 in the lemma above is called a **compact exhasution**. A locally compact Hausdorff space X having a compact exhasution is called **countable at infinity**. Hence, the Lemma above states that every open set in $\mathbb{C}^n$ is countable at infinity.

Define a topology τ_{co} on the space $\mathcal{C}(U, E)$ by the following family of seminorms

$$p_{k_j} : \mathcal{C}(U, E) \rightarrow \mathbb{R}$$

$$f \mapsto \sup_{x \in K_j} \|f(x)\|_E$$

where $(E, \|\cdot\|_E)$ is a Banach space. Then by functional analysis, we know that for a compact set K , the space $\mathcal{C}(U, E)$ is a Banach space with respect to the norm

$$\|f\|_{E, \infty} := \sup_{x \in K} \|f\|_E.$$

Proof: It's easy to see that $(\mathcal{C}(K, E), \|\cdot\|_{E, \infty})$ is a normed vector space. Let $(f_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in $\mathcal{C}(K, E)$, which means that

$$\|f_n - f_m\|_{E, \infty} = \sup_{x \in K} \|f_n(x) - f_m(x)\|_E \rightarrow 0 \quad (n, m \rightarrow \infty)$$

thus for any $\varepsilon > 0$, $\exists N_K \in \mathbb{N}$ such that for $\forall x \in K$ and $\forall n, m \geq N_K$, we have

$$\|f_n(x) - f_m(x)\|_E \leq \varepsilon$$

which shows that $(f_n(x))_{n \in \mathbb{N}}$ converges uniformly on E . Denote $f(x) := \lim_{n \rightarrow \infty} f_n(x)$, then $f \in \mathcal{C}(K, E)$ and

$$\|f_n - f\|_{E, \infty} = \sup_{x \in K} \|f_n(x) - f(x)\|_E \rightarrow 0 \quad (n \rightarrow \infty) \quad \square$$

Compact-Open Topology

Definition 2.1. : The topology τ_{co} on $\mathcal{C}(U, E)$ is called the **compact-open topology** or the **topology of compact convergence**.

Proposition 2.1. : A sequence $(f_j)_{j \in \mathbb{N}} \subseteq \mathcal{C}(U, E)$ converges with respect to the topology $\tau_{co} \Leftrightarrow (f_j)_{j \in \mathbb{N}}$ converges compactly on U .

Proof: Let $K \subseteq U$ be compact. By the lemma above, there is an index j_K such that $K \subseteq K_{j_K}$. If $f_j \rightarrow f$ with respect to the topology τ_{co} , then we know that

$$\sup_{x \in K} \|f_j(x) - f(x)\|_E = p_K(f_j - f) \leq p_{K_{j_K}}(f_j - f) \rightarrow 0,$$

which implies that $(f_j)_{j \in \mathbb{N}}$ converges uniformly on K ;

Conversely, if we have

$$\lim_{j \rightarrow \infty} \sup_{x \in K} \|f_j(x) - f(x)\|_E = 0$$

for any compact set $K \subseteq U$, then it is also true for the set K_j in Lemma 2.1. So by Lemma 1.2, we know that $f_j \rightarrow f$ with respect to the topology τ_{co} . $\square$

Proposition 2.2. : $(\mathcal{C}(U, E), \tau_{co})$ is complete.

Proof: Let $(f_j)_{j \in \mathbb{N}}$ be a Cauchy sequence in $\mathcal{C}(U, E)$ and $K \subseteq U$ is a compact set. Hence we know that $(f_j|_K)_{j \in \mathbb{N}}$ is a Cauchy sequence with respect to the norm $\|\cdot\|_{E, \infty}$. Since we have proved that $(\mathcal{C}(K, E), \|\cdot\|_{E, \infty})$ is a Banach space, so $\exists f^K \in \mathcal{C}(K, E)$ such that f^K is the uniform limit of $(f_j|_K)_{j \in \mathbb{N}}$. For $\forall z \in U$, $\exists$ an open neighborhood U_z of z such that the closure $\overline{U_z}$ is compact and $\overline{U_z} \subseteq U$. So by the discussion above, we have $\lim_{j \rightarrow \infty} f_j|_{\overline{U_z}} = f^{\overline{U_z}}$. Thus we can define

$$f(z) := \lim_{j \rightarrow \infty} f_j(z) \quad (\forall z \in U)$$

thus we can know that $f_j|_{\overline{U_z}}$ converges uniformly to $f|_{\overline{U_z}} = f^{\overline{U_z}} \in \mathcal{C}(\overline{U_z}, E)$, which implies that f is continuous at z . By the arbitrariness of z , we know that $f \in \mathcal{C}(U, E)$. $\square$

Remark: By the proof above, we can see that the only characteristic we need is the fact for any point $z \in U$, z has a compact neighborhood contained in U . Thus this proposition holds for the space $\mathcal{C}(X, E)$ of continuous mappings on any locally compact Hausdorff space X .

As a subspace of $\mathcal{C}(U, \mathbb{C}^m)$, $\mathcal{O}(U, \mathbb{C}^m)$ inherits the topology of compact convergence from $\mathcal{C}(U, \mathbb{C}^m)$.

Weierstrass

Theorem 2.1. : *Let $U \subseteq \mathbb{C}^n$ be an open set. Then $\mathcal{O}(U, \mathbb{C}^m)$ is a closed subspace of $\mathcal{C}(U, \mathbb{C}^m)$ with respect to the topology of compact convergence. For every $\alpha \in \mathbb{N}^n$, the linear operator*

$$D^\alpha : \mathcal{O}(U, \mathbb{C}^m) \rightarrow \mathcal{O}(U, \mathbb{C}^m)$$

$$f \mapsto D^\alpha f$$

is continuous. In case $m > 1$, i.e., $f = (f_1, f_2, \dots, f_m)$, the operator D^α has to be applied to every component function.

Proof: This assertion holds if and only if it holds true in each component, thus we may without loss of generality assume that $m = 1$.

Since $\mathcal{C}(U)$ is metrizable, it suffices to show that for any sequence $(f_j)_{j \in \mathbb{N}} \subset \mathcal{O}(U)$ converging compactly towards $f \in \mathcal{C}(U)$, we have $f \in \mathcal{O}(U)$ and the sequence $(D^\alpha f_j)_{j \in \mathbb{N}}$ converges compactly towards $D^\alpha f$:

For $\forall a \in U$, we have a polydisc such that $\overline{P_r^n(a)} \subseteq U$, hence the sequence $(f_j|_{\overline{P_r^n(a)}})_{j \in \mathbb{N}}$ converges uniformly towards $f|_{\overline{P_r^n(a)}}$. Let $T_r^n(a)$ be the polytorus contained in the boundary of $P_r^n(a)$ ($T_r^n(a) \subseteq \partial P_r^n(a)$). Then for $\forall z \in P_r^n(a)$, we have

$$\lim_{j \rightarrow \infty} \frac{f_j(w)}{(w - z)^1} = \frac{f(w)}{(w - z)^1} \text{ uniformly in } w \in T_r^n(a).$$

By CIF (Cauchy Integral Formula), we have

$$f(z) = \lim_{j \rightarrow \infty} f_j(z) = \lim_{j \rightarrow \infty} \left(\frac{1}{2\pi i} \right)^n \int_{T_r^n(a)} \frac{f_j(w)}{(w - z)^1} dw = \left(\frac{1}{2\pi i} \right)^n \int_{T_r^n(a)} \frac{f(w)}{(w - z)^1} dw.$$

By the previous discussion, we know that f is holomorphic. More over, by CIF, we have

$$D^\alpha f(z) = \alpha! \left(\frac{1}{2\pi i} \right)^n \int_{T_r^n(a)} \frac{f(w)}{(w - z)^{\alpha+1}} dw.$$

Let $K \subseteq U$ be compact, then there is a compact set $K' \subseteq U$ and a polytorus $r' \in \mathbb{R}_+^n$ such that $P_{r'}^n(a)$ is relatively compact in K' for all $a \in K$. By Cauchy Inequality, we have

$$\left| \frac{D^\alpha f(a)}{\alpha!} - \frac{D^\alpha f_j(a)}{\alpha!} \right| \leq \frac{1}{(r')^\alpha} \sup_{T_{r'}^n(a)} |f - f_j| \leq \frac{1}{(r')^\alpha} \sup_{K'} |f - f_j|.$$

Since f_j converges compactly to f and $a \in K$ is arbitrary, we know that $D^\alpha f_j$ converges to $D^\alpha f$ compactly on U , hence D^α is continuous. $\square$